

ThreadCare

A trusted bridge across every patient–clinician interaction

Technical Brief | Wang Yuyang | 27 August 2026 | Public synthetic-data prototype

Product thesis Medical history is not merely long; it mixes different writing styles, fragmented context, patient self-diagnosis, and information with unequal authority. ThreadCare lets AI compress that mess without becoming the source of truth: clinicians decide, and every AI claim remains traceable.

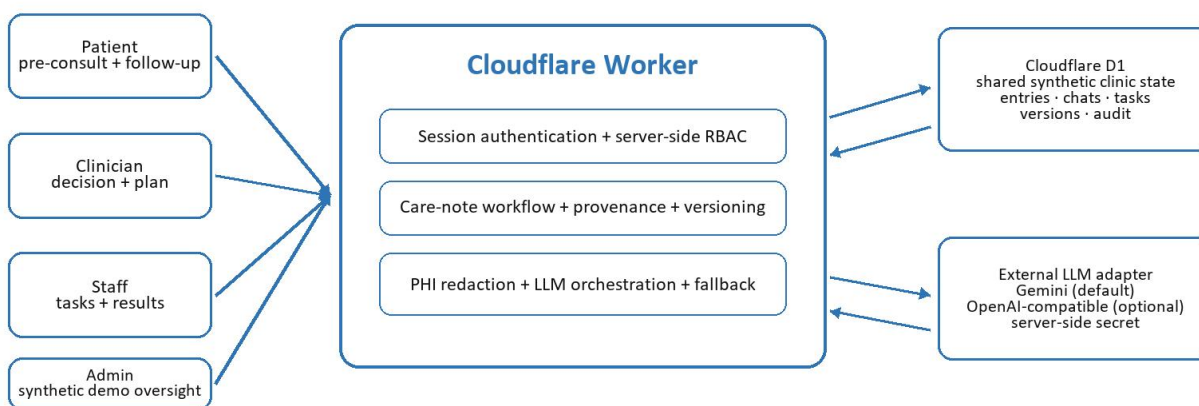
1. Problem, users, and design principles

The primary user is the care team. A clinician or staff member should efficiently understand the current concern, critical safety facts, and open actions, then move directly into the longitudinal evidence.

- The Top Card ranks a deliberately small set of actionable items; it does not present an AI suggestion as a confirmed fact.
- One shared longitudinal Care Note connects patient pre-consults, staff tasks/results, consultation outcomes, and patient-facing instructions. Additionally, clinicians can add an AI-generated summary of their conversation with the patient to the longitudinal timeline with a single click.
- The system provides a page where clinician can enter consultation outcomes and medication instructions, then simultaneously add them to the longitudinal timeline and send them to the patient.
- Role ownership is explicit: staff cannot overwrite clinician notes, clinicians cannot overwrite staff notes, and patients never receive internal comments or raw AI-scribed notes.
- After consultation, patients will be provided a form for rating and commenting. Thus the care team can get some feedback for further improvement.
- The prototype is a complete vertical slice at demo scale: one clinic, one clinician, two staff accounts, and multiple isolated patients rather than a feature-reduced mock-up.

2. Architecture

ROLE-SCOPED HTTPS CLIENTS



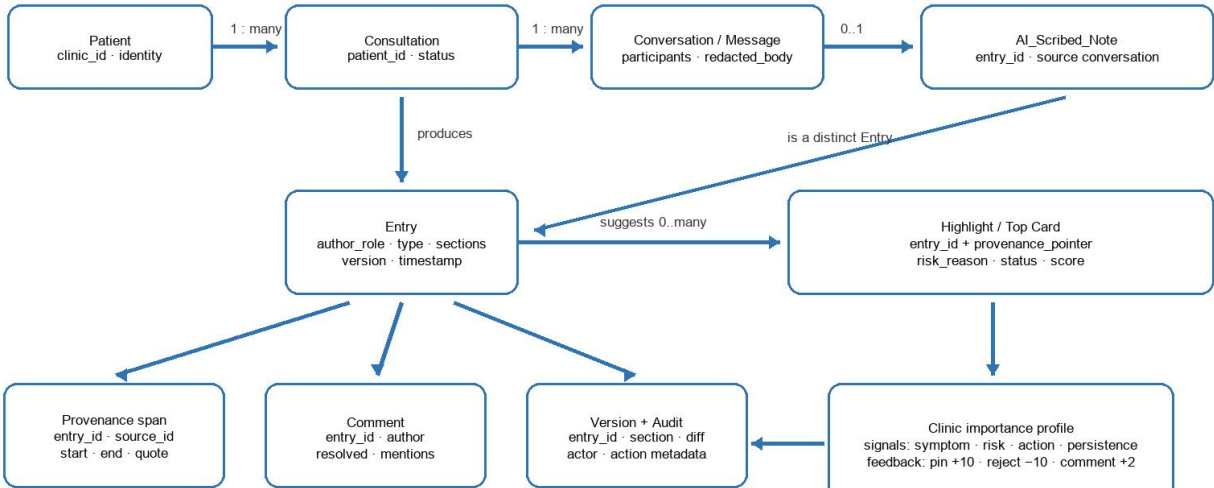
Public cross-device demo · HTTPS · synthetic data only · every summary remains linked to its exact source

Figure 1. Public HTTPS clients share one Worker-enforced, clinic-scoped D1 record; LLM providers never receive unredacted submitted text.

Public request path. Role-scoped browsers connect over HTTPS to a Cloudflare Worker. The Worker verifies the session, applies RBAC, and reads or mutates the shared synthetic clinic state in Cloudflare D1. Names, phone numbers, and supported IC/ID patterns are redacted before downstream summarisation. Gemini is the default LLM adapter, with an optional OpenAI-compatible fallback and deterministic fallbacks for temporary provider failure. Live demo: <https://threadcare-wang-yuyang.wang-yuyang.workers.dev>.

3. Data model: one record, preserved sources

Entry is the longitudinal unit. Manual notes and three AI-scribe types are visually distinct but use the same versioned entry structure. AI-scribed notes point to a complete source conversation; Top Card highlights additionally resolve to an exact entry/span.



Conflict rule: a clinician replacement supersedes an AI/patient assertion, while the original entry, source, and audit trail remain retrievable.

Figure 2. Logical schema and adaptive importance loop.

Concern	Implementation	Why it matters
Provenance	Source conversation/message + exact span pointers; original text remains retrievable.	A reviewer can inspect what the patient or team actually said, not only the AI's paraphrase.
Revision	Section-level versions, actor/timestamp metadata, audit log, deterministic same-section conflict response, and revert.	Concurrent edits to different sections survive; ownership and change history remain visible.
Conflict	Clinician replacement supersedes an AI/patient assertion while preserving the earlier claim and source.	Clinical authority wins without silently rewriting history.
Data decay	Old low-priority entries become archive cards; allergies, confirmed diagnoses, review items, unresolved tasks, comments, versions, and sources stay protected.	The default timeline becomes quieter without deletion or loss of traceability.

4. Self-learning

Ranking, not truth. Each suggestion combines model confidence/base score with explicit risk, unresolved actions, and clinic-level learned signals such as symptom, hypothesis, urgency, and category. A pin adds +10, a rejection -10, and a first team comment +2 to matching future signals. The score breakdown is inspectable; feedback never confirms a diagnosis, edits a source, or crosses clinic boundaries.

Hard constraint Only a clinician can accept or reject an AI highlight. Rejection records a reason and replacement assessment; the previous claim remains visible as superseded evidence.

5. Safety, permissions, and evidence

SERVER-SIDE GUARANTEES

- Role- and clinic-scoped session checks on protected routes
- Patient response excludes internal comments and raw AI sources
- Note edit/revert ownership enforced by author role
- Synthetic-only public demo; API keys stored as Worker secrets and excluded from Git
- Redaction before LLM: names, phones, supported IC/ID patterns

VERIFICATION

- 39 automated tests passing
- RBAC, revisions/revert, audit metadata, exact provenance
- Concurrent edits and deterministic same-section conflicts
- Multi-patient isolation and repeated consultations
- Self-learning transfer and non-destructive decay

Warm-path latency

Measured P95: 32.20 ms (requirement ≤ 300 ms). Measurement used 20 warm-up iterations followed by 200 sequential local samples of the authenticated Care Note and conversation-loading path before public deployment. The Cloudflare endpoint was separately smoke-tested for HTTPS, Gemini configuration, clinician login, and shared patient retrieval; no public-network P95 is claimed.

6. Assumptions and deliberate scope

Decision	Chosen scope	Trade-off / next production step
Voice	Reviewed speech-to-text input, maximum 30 seconds; raw audio is not retained.	Not full ambient capture: no diarisation, overlap/noisy-room handling, or continuous recording.
Storage	Cloudflare D1 persists one clinic-scoped JSON state snapshot for the public cross-device demo; local development uses an ignored JSON file.	Production needs a normalized transactional schema, encryption controls, migrations, backups, recovery, and concurrency testing.
Privacy	Public demo and public GitHub use synthetic data only.	Real PHI requires legal authority, security review, encryption at rest, retention policy, monitoring, and compliance controls.
Learning	Transparent clinic-level weighted feedback signals.	A production model needs evaluation for bias, drift, misuse, and safe rollback before learning affects workflow.

7. First-principles reflection

The key learning was understanding why clinical records become “messy.” Different people document different visits in different styles; histories accumulate without a shared summary; and a patient’s sincere self-diagnosis can be useful context while also distracting from the underlying condition. Therefore the core problem is authority and trust. ThreadCare treats AI as the clinician’s assistant: it organises and prioritises, while human clinical judgment decides what becomes accepted care context.